

Total No. of Questions : 6]

SEAT No. :

P60

OCT. -16/BE/Insem. - 114

[Total No. of Pages : 3

B.E. (Mechanical)

REFRIGERATION AND AIR CONDITIONING

(2012 Course) (402041) (Semester - I)

Time : 1Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) All questions carry equal marks.*
- 2) Use of non programmable scientific calculator is allowed.*
- 3) Assume suitable data if necessary.*

Q1) a) A refrigeration system based on Bell Coleman cycle working between the pressure limits of 4 bar and 16 bar extracts 125 MJ/h. The air enters the compressor at 5°C and enters the expander at 23°C. The compressor and expander efficiencies are 82% and 87% respectively. Determine: **[6]**

- i) power required to run the unit.
- ii) mass circulation rate of the air.
- iii) COP.
- iv) Carnot COP for same air entry temperature to compressor and expander.

Assume $C_p = 1.005 \text{ kJ/kg K}$, $C_v = 0.718 \text{ kJ/kg K}$.

b) With neat schematic, explain the refrigeration system for water cooler. **[4]**

OR

Q2) a) With the help of T-s diagram explain reversed Carnot cycle with vapour as refrigerant. Assuming unit mass circulation of refrigerant, derive the expression for COP of Carnot cycle. **[5]**

b) Explain the role of refrigeration system in dairy plant. **[5]**

P.T.O.

Q3) a) With T-s and P-h diagram explain the effect of evaporator pressure on COP of simple saturated vapor compression cycle. [4]

b) A simple saturated vapor compression cycle with R134-a refrigerant has temperature limits -10°C and 30°C . If the total load is 2 tons of refrigeration, using given properties,

Determine: [6]

- i) Condenser load kW,
- ii) COP
- iii) mass flow rate of refrigerant and
- iv) volume displacement of compressor.

OR

Q4) a) Explain with example use of: [4]

- i) Secondary refrigerants
- ii) Zeotropes and Azeotropes

b) A 5TR refrigerating machine based on R134-a vapor compression cycle operates between 12°C and 50°C . Temperature after sub-cooler is less by 5 degree. Use data given. Determine: [6]

- i) COP.
- ii) Power per ton of refrigeration
- iii) Mass circulation of refrigerant in kg/h

Q5) a) Explain two stage compressions with intercooling and sub cooling by external cooling source. [4]

b) Draw a neat diagram of 'Electrolux refrigerator' and explain its working principle. What is the role of hydrogen in this refrigeration system? [6]

OR

Q6) a) What is the basic function of a compressor in a vapor compressor refrigeration system? How this function is achieved in vapor absorption system? [4]

b) A data pertaining to refrigeration system is as follows: [6]

Two separate evaporators E_1 (10 TR) and E_2 (20 TR) at temperatures 5°C and 15°C respectively. Common condenser is at a temperature 38°C . The liquid from the condenser is saturated. After evaporator E_2 , pressure reducing valve bring the pressure to the suction pressure of the compressor. Single compressor and individual expansion valves are used in the system. Use Refrigerant data provided.

Make the neat flow diagram and a cycle diagram on P-h plane, Find:

- Power required to run the system.
- COP of the system.

Refrigerant Data Q6								
T	P	v_f	v_g	h_f	h_g	h_{fg}	s_f	s_g
$^\circ\text{C}$	bar	m^3/kg		kJ/kg			kJ/kgK	
5	3.63	0.000725	0.0475	40.69	189.65	148.96	0.1586	0.6943
15	4.91	0.000743	0.0355	50.10	193.79	143.69	0.1915	0.6902
38	9.14	0.000794	0.0191	72.56	202.51	129.95	0.2654	0.6830

R134-a Data								
T	P	v_g	h_f	h_g	s_f	s_g	C_{pliq}	C_{pvap}
$^\circ\text{C}$	bar	m^3/kg	kJ/kg	kJ/kg	kJ/kgK	kJ/kgK	kJ/kgK	kJ/kgK
-10	2.0052	0.09963	186.78	--	0.9509	1.7337	--	0.842
-6	2.3418	0.08591	192.03	395.15	0.9707	1.7310	--	0.858
2	3.145	0.0647	202.68	399.84	--	1.7263	1.341	0.892
12	4.4289	0.04636	--	405.51	1.0579	1.7215	1.374	0.939
24	6.4566	0.03189	233.05	411.93	--	1.7169	--	1.006
30	7.7008	0.02667	241.65	414.94	1.1432	1.7149	--	1.044
50	13.177	--	271.59	423.63	1.2373	1.7078	1.569	1.218

